

HYDROGENATED NBR: PATENT LANDSCAPE ON HNBR SYNTHESIS, HYDROGENATION, AND PROCESS/COMPOSITION CONTROL

1. ABSTRACT

This report synthesizes a patent landscape centered on hydrogenated nitrile rubber (HNBR) and closely related process technologies. The target is defined as nitrile-butadiene rubber precursors and their hydrogenated, partially hydrogenated, or fully hydrogenated derivatives, with emphasis on synthesis, hydrogenation, residual unsaturation control, molecular-weight control, and manufacturing robustness. Across the selected patents, the dominant technical theme is solution-phase hydrogenation of NBR using homogeneous noble-metal catalysts, especially ruthenium-, rhodium-, and palladium-based systems, with strong attention to catalyst loading, hydrogen pressure, temperature, and post-reaction catalyst removal. One key route (EP2473281B1) combines metathesis and hydrogenation in a single process to reduce molecular weight while hydrogenating the polymer, using ruthenium/osmium complexes and hydrogen in the presence of optional co-olefin. Another route (EP3827029A1) focuses on hydrogenating NBR in solution with controlled noble-metal content and operating windows designed to improve ageing properties while minimizing catalyst residues. EP2292687B1 is not a synthesis patent, but it is highly relevant because it discloses HNBR-based vulcanizable compositions and explicitly defines HNBR precursor and hydrogenation characteristics, including hydrogenation degree, Mooney viscosity, and cure-package design. US11131058B2 likewise is formulation/process-oriented, showing how carboxy-modified NBR/HNBR latex systems are combined with resorcinol-formaldehyde and polycarbodiimide chemistry for cord treatment, indirectly reflecting HNBR compatibility in industrial rubber systems. The remaining patents are less directly synthetic but still informative: US9321910B2 and US10385647B2 show HNBR in demanding end-use compositions, while EP2519350B1 and EP3137587B1 are largely outside the HNBR synthesis domain. Overall, the landscape indicates that practical HNBR manufacture depends on balancing hydrogenation efficiency, catalyst economy, residual metal control, and preservation of polymer properties, while downstream formulations exploit HNBR's heat and oil resistance in high-temperature sealing, belt, and swellable-rubber applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a one-pot process for producing low-molecular-weight hydrogenated nitrile rubber by reacting nitrile rubber in the presence of hydrogen and a ruthenium- or osmium-containing metathesis/hydrogenation catalyst. The process is designed to combine metathesis-driven molecular-weight reduction with hydrogenation, optionally in the presence of a co-olefin, to obtain HNBR with lower molecular weight and narrower molecular-weight distribution. - Catalyst metal: ruthenium or osmium — Claims - Hydrogen presence: in the presence of hydrogen — Claims - Optional co-olefin: at least one co-olefin — Claims - Molecular weight range: 20,000 to 250,000 — Description - Mooney viscosity (ML 1+4 @ 100°C): 1 to 50 — Description - Polydispersity index (MWD): low / narrower than prior art — Description - Catalyst amount: 0.041 phr — Example 1, HNBR 1 - Catalyst amount: 0.045 phr — Example 1, HNBR 2 - Catalyst amount: 0.060 phr — Example 1, HNBR 3 - Metal content: 65.6 ppm — Example 1 - Co-catalyst: triphenylphosphine — Example 1, HNBR 3 - Co-catalyst amount: 1.0 phr — Example 1, HNBR 3 - Reaction time: 0, 60, 120, 180, 240 min — Example 1 - Residual double bond content (RDB): 100% at 0 min; 2.5%, 7.0%, 15.4% at 60 min; 0.6%, 4.5%, 3.1% at 120 min; 0.5%, 3.9%, 1.3% at 180 min; 0.4%, 3.5%, 0.7% at 240 min — Example 1 - Mw: 113371 g/mol — Example 1, HNBR 1 - Mn: 48352 g/mol — Example 1, HNBR 1 - PDI (Mw/Mn): 2.34 — Example 1, HNBR 1 - Mooney viscosity: 33.3 — Example 1, HNBR 1 - Mw: 133758 g/mol — Example 1, HNBR 2 - Mn: 54595 g/mol — Example 1, HNBR 2 - PDI (Mw/Mn): 2.45 — Example 1, HNBR 2 - Mooney viscosity: 41.3 — Example 1, HNBR 2 - Mw: 195304 g/mol — Example 1, HNBR 3 - Mn: 64130 g/mol — Example 1, HNBR 3 - PDI (Mw/Mn): 3.05 — Example 1, HNBR 3 - Mooney viscosity: 69.3 — Example 1, HNBR 3

Relevant Experimental Evidence - Example 1: Compared three catalyst systems and showed that the inventive catalyst gave faster hydrogenation and lower residual double bond content than the industrial benchmark. - Example 1: Demonstrated concurrent molecular-weight reduction during hydrogenation, yielding low-Mw HNBR with reduced Mooney viscosity. - Example 1: Showed that catalyst choice strongly affected hydrogenation profile, residual unsaturation, and final polymer rheology.

Technical Relevance This is one of the most directly relevant patents because it explicitly targets HNBR synthesis and combines hydrogenation with molecular-weight control. It provides concrete operating and product-property evidence for low-residual-unsaturation HNBR manufacture.

Patent 2

- Patent Number: US9321910B2
- Patent Title: Microgel-containing vulcanizable composition based on hydrogenated nitrile ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not a synthesis route for HNBR itself; it discloses vulcanizable compositions based on hydrogenated nitrile rubber combined with unsaturated carboxylic acids or salts, peroxide, and microgels. The HNBR is used as a high-temperature elastomeric base in cured articles, with emphasis on improved mechanical properties and processing behavior. - Hydrogenated nitrile rubbers: 100 parts by weight based on 100 parts by weight of rubber — Claim 1 - Mooney viscosity (ML 1+4 @ 100°C): 10 to 120 MU — Claim 1 - Degree of hydrogenation of diene units: 95 to 100% — Claim 1 - Unsaturated carboxylic acids/salts: 1 to 100 parts by weight — Claim 1 - Peroxide: 0.2 to 8 parts by weight — Claim 1 - Microgel glass transition temperature: less than or equal to -20°C — Claim 1 - Microgel amount: up to 60 parts by weight — Claim 2 - Microgel glass transition temperature: less than or equal to -50°C — Claim 3 - Mooney viscosity (ML 1+4 @ 100°C): 15 to 100 MU — Claim 4 - HNBR1 residual double bond content: 0.4% — Example 2 - HNBR1 maleic monoethylester content: 4.5% — Example 2 - HNBR1 Mooney viscosity (ML 1+4 @100°C): 101 — Example 2 - NBR1 butadiene content: 60.4% by weight — Example 2 - NBR1 termonomer content: 4.5% by weight — Example 2 - NBR1 acrylonitrile content: 35.1% by weight — Example 2 - NBR1 Mooney viscosity (ML 1+4 @100°C): 35 — Example 2 - NBR solution concentration: 12% total solids in monochlorobenzene — Example 2 - Reaction temperature: 138°C — Example 2 - Agitation speed: 600 rpm — Example 2 - Hydrogen pressure: 85 bar — Example 2 - Reaction time: 4 hrs — Example 2 - Cure temperature: 180°C — Example 1 - Press-cure pressure: 100 bar — Example 1 - Post-cure temperature: 175°C — Example 1 - Post-cure time: 4 hours — Example 1

Relevant Experimental Evidence - Example 1: Showed that microgel-containing HNBR vulcanizates can be press-cured and post-cured to deliver improved high-temperature mechanical properties. - Example 2: Demonstrated solution hydrogenation of NBR1 to HNBR1 using Wilkinson's catalyst and triphenylphosphine at 138°C and 85 bar H₂. - Example 2: Reported that the resulting HNBR1 had very low residual double bond content (0.4%) and high Mooney viscosity (101).

Technical Relevance Although focused on formulations rather than polymer synthesis, this patent is highly relevant because it defines HNBR precursor characteristics and hydrogenation degree used in high-performance applications. It also provides a practical benchmark for HNBR properties and downstream cure systems.

Patent 3

- Patent Number: EP2519350B1
- Patent Title: Highly active metathesis catalysis selective for romp and rcm
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses novel ruthenium metathesis catalysts and their preparation, with the stated utility of carrying out RCM, CM, ADMET, and ROMP reactions. It also mentions use in depolymerization of NBR/SBR and in hydrogenation-related rubber processing, but the examples primarily concern catalyst synthesis rather than HNBR production. - Ligand synthesis scale: 5.0 mmol — Example 1 - DME: 10 mL — Example 1 - Deionized water: 3 mL — Example 1 - K₂CO₃: 1.5 eq — Example 1 - Reaction temperature: 85°C — Example 1 - 2,4,6-Trivinyl-cyclotriboroxane pyridine complex: 0.5 eq — Example 1 - Pd(PPh₃)₄: 2% — Example 1 - Ligand yield: 86% — Example 1 - Ru complex synthesis scale: 1.0 mmol — Example 1 - CuCl: 270 mg, 2.5 mmol, 2.5 eq — Example 1 - DCM: 15 mL — Example 1 - Ligand 3a: 250 mg, 1.2 mmol, 1.2 eq — Example 1 - Reaction time: 30-60 min — Example 1 - Ru complex yield: 4% — Example 1 - Ligand 3b yield: 91% — Example 2 - Ru complex 4b yield: 29% — Example 2 - Ligand 3c yield: 54% — Example 3 - Ru complex 4c yield: 5% — Example 3 - Ligand 3d yield: 62% — Example 4 - Ru complex 4d yield: 32% — Example 4 - Ru complex 4e yield: 35% — Example 5 - Ligand 3f yield: 63% — Example 6 - Ru complex 4f yield: 14% — Example 6 - Ligand 3g yield: 56% — Example 7 - Ru complex 4g yield: 8% — Example 7 - Ligand 3h yield: 32% — Example 8 - Ru complex 4h yield: 42% — Example 8 - Ligand 3j yield: 93% — Example 9 - Ru complex 4j yield: 48% — Example 9 - Ligand 3k yield: 83% — Example 10 - Ru complex 4k yield: 68% — Example 10 - Ligand 3m yield: 49% — Example 11 - Ru complex 4m yield: 52% — Example 11 - Ligand 3n yield: 45% — Example 12 - Ru complex 4n yield: 64% — Example 12 - Ligand 3p yield: 67% — Example 13 - Ru complex 4p yield: 26% — Example 13 - Ligand 3q yield: 46% — Example 14 - Ru complex 4q yield: 38% — Example 14 - Ligand 3r yield: 44% — Example 15 - Ru complex 4r yield: 14% — Example 15 - Ligand 3s yield: 51% — Example 16 - Ru complex 4s yield: 85% — Example 16 - Ligand 3t yield: 38% — Example 17 - Ru complex 4t yield: 23% — Example 17 - Ligand 3u yield: 28% — Example 18 - Ru complex 4u yield: 32% — Example 18 - Ligand 3v yield: 81% — Example 19 - Ru complex 4v yield: 10% — Example 19 - Ru complex 4w yield: 31% — Example 20 - Depolymerization temperature: 30-100°C — Description - Hydrogenation temperature: 60-150°C — Description - Hydrogen under high pressure: present — Description

Relevant Experimental Evidence - Example 1: Prepared a new Ru complex from a ligand precursor and showed catalyst formation under inert atmosphere. - Example 2: Demonstrated an alternative Ru complex synthesis route with a different precursor and lower isolated yield. -

Examples 1-20: Established a library of Ru metathesis catalysts with varying ligand structures and isolated yields.

Technical Relevance This patent is relevant mainly as a catalyst-enabling disclosure rather than a direct HNBR synthesis route. It is useful because it explicitly mentions depolymerization of NBR and hydrogenation of HNBR under high-pressure hydrogen, indicating catalyst concepts applicable to HNBR processing.

Patent 4

- Patent Number: EP2292687B1
- Patent Title: ... containing ethylene acrylate (AEM), polyacrylate (ACM) and/or hydrogenated ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses guanidine-free vulcanizable mixtures based on AEM, ACM, and/or HNBR rubbers, using non-aromatic amine or amidine accelerators together with retarder systems such as benzoic acid or carbodiimides. It is a formulation patent, but it explicitly defines HNBR composition, hydrogenation level, and solution hydrogenation conditions used to prepare the HNBR precursor. - HNBR Mooney viscosity (ML 1+4 @ 100°C): 101 — Example 2 - Residual double bond content: 0.4% — Example 2 - Maleic monoethylester content: 4.5% — Example 2 - NBR1 butadiene content: 60.4% by weight — Example 2 - NBR1 termonomer content: 4.5% — Example 2 - NBR1 acrylonitrile content: 35.1% by weight — Example 2 - NBR1 Mooney viscosity (ML 1+4 @100°C): 35 — Example 2 - NBR solution concentration: 12% total solids — Example 2 - Solvent: monochlorobenzene — Example 2 - Reaction temperature: 138°C — Example 2 - Agitation speed: 600 rpm — Example 2 - Hydrogen pressure: 85 bar — Example 2 - Reaction time: 4 hrs — Example 2 - Hydrogenated nitrile rubbers: degree of hydrogenation 95 to 100% — Claim 1 - Hydrogenated nitrile rubbers: Mooney viscosity (ML 1+4 @ 100°C) 10 to 120 MU — Claim 1 - Unsaturated carboxylic acids/salts: 1 to 100 parts by weight — Claim 1 - Peroxide: 0.2 to 8 parts by weight — Claim 1 - Microgel glass transition temperature: less than or equal to -20°C — Claim 1 - Microgel amount: up to 60 parts by weight — Claim 2 - Stabaxol I: 1.0 phr — Example 2 - DIAK NO.1: 2.5 phr — Example 2 - Corax N-550: 40.0 phr — Example 2 - Total phr: 153.5 to 155.0 — Example 2 - MDR test temperature: 180°C for 30' — Example 2 - t50: 125, 133, 136 s — Example 2 - t90: 382, 434, 453 s — Example 2 - t95: 489, 590, 634 s — Example 2

Relevant Experimental Evidence - Example 1: Compared guanidine-containing and guanidine-free cure packages for HNBR-related vulcanizable mixtures. - Example 2: Showed solution hydrogenation of NBR1 to HNBR1 at 138°C and 85 bar H₂, yielding 0.4% residual double

bonds. - Example 2: Demonstrated that retarder addition shifts vulcanization times to longer values, improving scorch safety.

Technical Relevance This patent is relevant because it ties HNBR precursor preparation to practical vulcanization chemistry and explicitly documents a solution hydrogenation route. It also provides formulation constraints that matter for converting HNBR into usable rubber goods.

Patent 5

- Patent Number: US10385647B2
- Patent Title: Swellable rubber compositions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses swellable rubber compositions for downhole packers, using nitrile elastomer or hydrogenated nitrile elastomer together with butadiene rubber, sulfur curing agent, silica, and powder super absorbent polymer. It is not a synthesis patent for HNBR, but it shows how HNBR is incorporated into high-temperature swellable elastomer systems. - Powder super absorbent polymer: about 125 Phr to about 175 Phr — Claim 2 - Powder super absorbent polymer: about 170 Phr to about 210 Phr — Claim 3 - Powder super absorbent polymer: at least about 150 Phr, 175 Phr, 200 Phr or 225 Phr — Claim 4 - Total super absorbent polymer fraction: at least about 50%, 60%, 70%, 80%, 90%, or 95% by weight — Claim 5 - Hydrogenated nitrile elastomer hydrogenation: about 70 percent to about 91 percent — Description - Elastomer Mooney viscosity ML 1+4 (100°C): about 47 to about 72 — Description - Hydrogenated nitrile elastomer unsaturation: about 9 percent to about 30 percent — Description - Powder SAP loading: 205 Phr — Example 4 - NBR rubber loading: 10% — Example 4 - HNBR loading: 10% — Example 4 - 200°F, 5% brine, 96 hours: 75.8% swell for DN3350 composition — Example 4 - 200°F, 16% brine, 96 hours: 46.5% swell for DN3350 composition — Example 4 - 275°F, 5% brine, 96 hours: 98.4% swell for DN3350 composition — Example 4 - 275°F, 16% brine, 96 hours: 65.4% swell for DN3350 composition — Example 4 - 200°F, 5% brine, 96 hours: 77.8% swell for HNBR composition — Example 4 - 200°F, 16% brine, 96 hours: 47.2% swell for HNBR composition — Example 4 - 275°F, 5% brine, 96 hours: 96.0% swell for HNBR composition — Example 4 - 275°F, 16% brine, 96 hours: 66.9% swell for HNBR composition — Example 4 - Cure temperature: 330°F — Example 1 - Cure pressure: 70 psi — Example 1 - Cure time: 2.5 hours — Example 1 - Downhole test temperature: 275°F — Example 1 - Downhole pressure: about 300 psi — Example 1 - Differential pressure: at least 5000 psi — Example 1

Relevant Experimental Evidence - Example 1: Demonstrated that a swellable packer cured at 330°F and 70 psi could withstand 275°F brine exposure and >5000 psi differential pressure. -

Example 2: Compared powder, medium granular, and large granular SAPs across multiple loadings and showed strong dependence of swell and leaching on particle size. - Example 4: Showed that replacing NBR with HNBR at 10% maintained or improved swell performance under hot brine conditions.

Technical Relevance This patent is relevant because it shows HNBR used in a demanding elastomeric application where hydrogenation level and thermal stability matter. It provides practical evidence that HNBR can improve performance in high-temperature, high-pressure swellable systems.

Patent 6

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a method for producing HNBR by hydrogenating NBR in solution using ruthenium, palladium, or rhodium compounds, with emphasis on catalyst recovery and minimizing residual noble metals. The process is framed around achieving good ageing properties while controlling catalyst loading, temperature, pressure, and reaction time. - NBR repeat units: 40% to 90% by weight conjugated diene — Claim 1 - NBR repeat units: 10% to 60% by weight α,β -unsaturated nitrile — Claim 1 - Ruthenium amount: 10 ppm to 200 ppm — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200°C — Claim 1 - Hydrogenation pressure: 700000 pascals to 15000000 pascals — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1 - Ageing test: 4 days at 140°C — Description - Gel content increase during ageing: less than 10% — Description - Mooney viscosity increase during ageing: less than 40% — Description

Relevant Experimental Evidence - No numbered examples were detected by the parser; the disclosure is primarily claim- and description-driven. - The patent emphasizes catalyst residue control and ageing performance rather than example-by-example synthesis data. - It defines broad operating windows for hydrogenation of NBR to HNBR with Ru, Pd, or Rh catalysts.

Technical Relevance This is one of the most directly relevant patents because it explicitly claims HNBR production from NBR in solution and gives practical catalyst and process windows. Its focus on residual metal control and ageing stability is highly pertinent to industrial HNBR manufacture.

Patent 7

- Patent Number: EP3137587B1
- Patent Title: Detergent composition
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not about HNBR synthesis; it concerns detergent compositions containing DNase for biofilm removal. It has no technical connection to hydrogenated nitrile rubber synthesis or processing. - DNase concentration: 10 ppm — Example 1 - Incubation temperature: 30°C — Example 1 - Incubation time: 1 day — Example 1 - Centrifugation: 3000 g at 21°C for 7 min — Example 1 - Biofilm incubation: 72 h at 37°C — Example 1 - Crystal violet: 0.1% (w/v) — Example 1 - Ethanol: 96% (w/v) — Example 1 - DNase concentration: 0.1 ppm — Example 2 - DNase concentration: 0.3 ppm — Example 2 - Wash liquor temperature: 30°C — Example 2 - Wash liquor time: 1 hour — Example 2 - Shaking speed: 800 rpm — Example 2 - DNase concentration: 0.03 ppm — Example 3 - DNase concentration: 0.1 ppm — Example 3 - DNase concentration: 0.3 ppm — Example 3 - DNase concentration: 1 ppm — Example 3 - Incubation temperature: 15°C — Example 3 - Incubation time: 2 days — Example 3 - Treatment temperature: 45°C — Example 4 - Treatment time: 30 and 60 min — Example 4

Relevant Experimental Evidence - Example 1: Demonstrated DNase-mediated reduction of biofilm adhesion on multiple hard surfaces. - Example 2: Showed reduced stickiness of biofilm on plastic surface with 0.1 and 0.3 ppm DNase. - Example 3: Demonstrated biofilm removal from plastic surface across 0.03 to 1 ppm DNase. - Example 4: Compared detergent treatments on biofilm-forming bacteria under 45°C washing conditions.

Technical Relevance This patent is not technically relevant to HNBR synthesis. It does not disclose hydrogenated nitrile rubber chemistry, hydrogenation, or rubber formulation concepts applicable to HNBR manufacture.

Patent 8

- Patent Number: US11131058B2
- Patent Title: Method for manufacturing core wire for transmission belt, treatment agent, and ...
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a method for manufacturing a cord for a power transmission belt using a first treatment agent containing resorcin-formaldehyde condensate, a carboxy-modified latex, and a polycarbodiimide curing agent in a hydrophilic solvent. It is not an HNBR synthesis patent, but it explicitly includes carboxy-modified hydrogenated acrylonitrile-butadiene copolymer latex among the usable latex types. - Polycarbodiimide resin NCN equivalent: 300 to 450 — Claim 1 - Condensate (A1): 10 to 100 parts by mass relative to 100 parts by mass of rubber component (A2) — Claim 1 - Curing agent (A3): 1 to 10 parts by mass relative to 100 parts by mass of rubber component (A2) — Claim 2 - Rubber composition (A): 15% to 20% by mass of first treatment agent — Description - Hydrophilic solvent (B): water — Claim 1 - First treatment agent solid content: 17% to 20% by mass — Example 5 / Example 9 tables - RFL-X blend amount: 625 g — Example 5 - RF condensate: 25 g — Example 5 - Polycarbodiimide dispersion liquid A: 12.5 g, 25 g — Example 5 - Polycarbodiimide dispersion liquid B: 2.5 g, 12.5 g — Example 5 - Blocked isocyanate dispersion liquid: 33.3 g — Example 5 - Water: 54.2 g to 97.5 g — Example 5 - Total solid content concentration: 17% to 19% — Example 5 - EPDM composition: 100 parts by mass — Example 5 - Stearic acid: 5 parts by mass — Example 5 - Zinc oxide: 5 parts by mass — Example 5 - HAF carbon black: 10 parts by mass — Example 5 - Hydrous silica: 15 parts by mass — Example 5 - Vulcanization accelerator MBTS: 2 parts by mass — Example 5 - Sulfur: 1 part by mass — Example 5 - Twist factor: 0.01 to 10 — Description - Number of twist: 20 to 300 times/m — Description

Relevant Experimental Evidence - Example 5: Showed multiple treatment-agent formulations using carboxy-modified NBR and polycarbodiimide dispersion liquids with total solid contents of 17-19%. - Example 9: Compared treatment-agent formulations with different solid contents and polymeric additives for cord adhesion. - The disclosure indicates that carboxy-modified hydrogenated acrylonitrile-butadiene copolymer latex is a permissible component in the treatment system.

Technical Relevance This patent is only indirectly relevant to HNBR because it includes carboxy-modified HNBR latex as a possible adhesive component. It does not teach HNBR synthesis, but it does show an industrial use context where hydrogenated nitrile rubber chemistry is compatible with cord-treatment formulations.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2473281B1 and EP3827029A1 are the most synthesis-focused HNBR patents: EP2473281B1 emphasizes simultaneous metathesis/hydrogenation for molecular-weight reduction, while EP3827029A1 emphasizes conventional solution hydrogenation with controlled noble-metal loading and catalyst removal.
- EP2473281B1 discloses very low catalyst loadings in phr and tracks residual double bond content over 0-240 min, whereas EP3827029A1 defines catalyst content in ppm, hydrogenation at 60-200°C, 700000-15000000 Pa, and 1-24 h.
- US9321910B2 and EP2292687B1 are formulation patents rather than synthesis patents; both define HNBR as a high-hydrogenation elastomer and provide cure-package/process windows, but neither introduces a new hydrogenation route.
- US10385647B2 uses HNBR in swellable packers and highlights hydrogenation levels of about 70-91% and unsaturation of about 9-30%, showing a less fully hydrogenated elastomer than the 95-100% hydrogenation claimed in EP2292687B1.
- US11131058B2 is the most indirect HNBR reference: it does not synthesize HNBR, but it explicitly allows carboxy-modified hydrogenated acrylonitrile-butadiene copolymer latex in a cord-treatment system.
- EP2519350B1 is catalyst-centric and only indirectly relevant to HNBR; it mentions depolymerization of NBR and hydrogenation of HNBR at 60-150°C, but the examples are focused on Ru-complex synthesis rather than rubber hydrogenation.
- EP3137587B1 is outside the HNBR technical field and serves as a non-relevant control patent in this manifest.
- Across the relevant patents, the recurring process themes are solution-phase hydrogenation, noble-metal catalyst management, residual unsaturation control, and downstream formulation compatibility for high-temperature rubber goods.

4. CONCLUSION

The evidence shows that HNBR manufacturing is dominated by solution-phase hydrogenation of NBR using noble-metal catalysts, with two major technical directions: simultaneous metathesis/hydrogenation for molecular-weight reduction (EP2473281B1) and conventional hydrogenation with tight catalyst-residue control (EP3827029A1). For practical HNBR production, the strongest evidence supports hydrogenation in the range of 60-200°C and 700000-15000000 Pa, with catalyst contents in the tens of ppm and reaction times of 1-24 h, while aiming for very low residual double bond content and minimal catalyst carryover. For high-performance end use, the formulation patents indicate that highly hydrogenated HNBR, typically 95-100% hydrogenation, is preferred for heat- and oil-resistant vulcanizates, and that cure packages, microgels, and retarder systems can be used to tune processing and final properties. Recommended synthesis parameters based on the evidence are: solution hydrogenation of NBR with ruthenium/rhodium/palladium catalysts in the low-ppm range, operation near 138°C and 85 bar where demonstrated, and post-reaction catalyst removal or minimization to preserve ageing performance. Where molecular-weight reduction is desired, the simultaneous metathesis/hydrogenation route is the most distinctive disclosed option.

5. REFERENCES

- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US9321910B2 | Microgel-containing vulcanizable composition based on hydrogenated nitrile ... | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9321910B2/en>
- EP2519350B1 | Highly active metathesis catalysis selective for romp and rcm | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2519350B1/en>
- EP2292687B1 | ... containing ethylene acrylate (AEM), polyacrylate (ACM) and/or hydrogenated ... | Not disclosed | EP | 2012 | <https://patents.google.com/patent/EP2292687B1/en>
- US10385647B2 | Swellable rubber compositions | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10385647B2/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>
- EP3137587B1 | Detergent composition | Not disclosed | EP | 2018 | <https://patents.google.com/patent/EP3137587B1/en>
- US11131058B2 | Method for manufacturing core wire for transmission belt, treatment agent, and ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11131058B2/en>